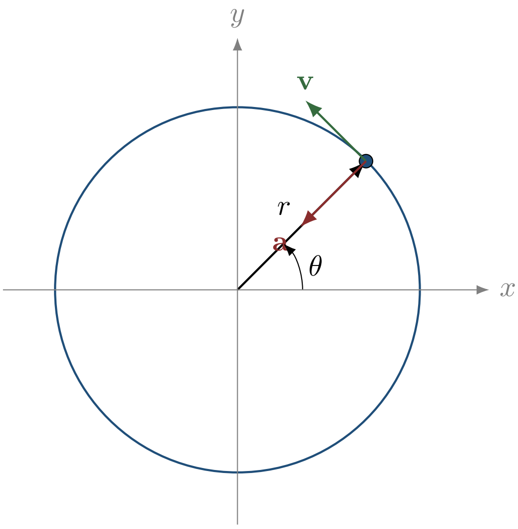
Circular Motion

A Narrative Route from Kinematics to Forces, Orbits, Magnetic Motion, and the Bohr Atom



Lecture notes for students who must learn to read equations as compressed physical information, not as a collection of formulas to memorize.

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How to read these notes

This document continues the same idea as the projectile-motion recap: physics is not a catalogue of formulas. A formula is a compact way to encode relations between quantities. The important skill is not to ask first, "Which formula should I use?" The important skill is to ask:

What is the physical situation, what is the mathematical model, and what condition answers the question?

Circular motion is an excellent place to practise this skill because the same mathematical structure appears in many physical systems:

- a stone tied to a string,
- a passenger on a carousel,
- a car turning on a road,
- a satellite orbiting a planet,
- two charges attracting each other,
- a charged particle moving in a magnetic field,
- the elementary Bohr model of the atom.

The surface situation changes, but the core question often remains the same:

> *If the path is circular, what force must point toward the centre, and what relation does that impose between radius, speed, angular velocity, period, mass, charge, field strength, or gravitational parameter?*

The notes are deliberately detailed. Many intermediate steps are written out, because the goal is to show the movement of thought:

$$\textbf{geometry} \longrightarrow \textbf{kinematics} \longrightarrow \textbf{dynamics} \longrightarrow \textbf{physical interpretation}.$$

This movement of thought is more important than any single final expression. A student who only memorizes $a_c = v^2/R$ may still be helpless when the problem is written in different words. A student who understands why a changing velocity direction requires inward acceleration can rebuild the formula, choose the correct radius, and identify the physical force that supplies the acceleration. This is the difference between using mathematics as a language and using mathematics as a box of disconnected recipes.

For that reason, many calculations below are accompanied by verbal pauses. These pauses are not decoration. They are meant to model the internal questions that a student should learn to ask before, during, and after the algebra.

The central message

Circular motion is not a special formula. It is a structure. Once we understand the structure, many apparently different problems become variations of the same mathematical idea.

1 The physical idea before the equations

Imagine a particle moving along a circle. Its distance from the centre is constant. The position changes, but the radius does not. This already tells us something important: the velocity cannot point along the radius. If the velocity had a radial component, the distance from the centre would change. Therefore, for motion exactly along a circle, the velocity must be tangent to the circle.

That is a physical sentence. In mathematics we will encode it by saying that the position vector has constant length and that its derivative is perpendicular to the position vector. This is the first lesson:

"The distance from the centre is constant" becomes an equation.

Let the centre of the circle be the origin. The position vector is

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}.$$

The particle moves on a circle of radius R if

$$|\mathbf{r}(t)| = R.$$

Equivalently,

$$x(t)^2 + y(t)^2 = R^2.$$

This equation is not a formula for numerical substitution. It is a statement about the geometry of the motion.

1.1 From constant distance to perpendicular velocity

Differentiate the constraint

$$x(t)^2 + y(t)^2 = R^2.$$

The right-hand side is constant, so its derivative is zero:

$$2x(t)x'(t) + 2y(t)y'(t) = 0.$$

Divide by 2:

$$x(t)v_x(t) + y(t)v_y(t) = 0.$$

In vector notation this is

$$\mathbf{r}(t) \cdot \mathbf{v}(t) = 0.$$

This is a precise mathematical expression of a geometric fact:

the radius vector is perpendicular to the velocity vector.

Translation of the first idea

Physical statement	Mathematical encoding	Meaning
The particle stays on a circle.	$x^2 + y^2 = R^2$	distance from centre is constant
The velocity is tangent.	$\mathbf{r} \cdot \mathbf{v} = 0$	no radial component of velocity
The direction of motion changes.	$\mathbf{a} \neq 0$ may hold even if $\lvert\mathbf{v}\rvert$ is constant	speed and velocity are not the same

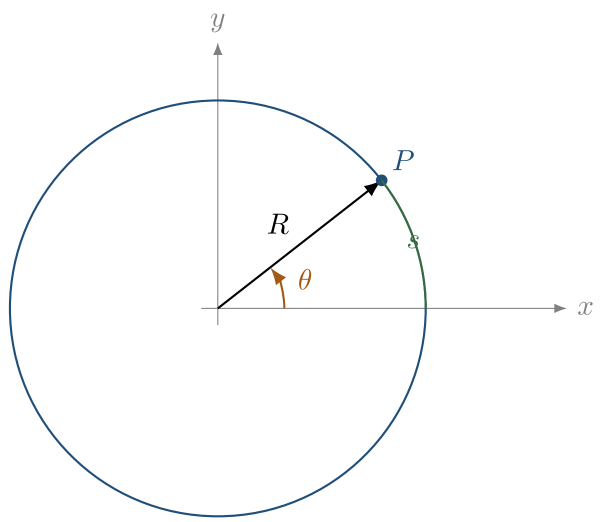
This last line is crucial. A body can move with constant speed and still accelerate. Acceleration measures change of velocity, and velocity has both magnitude and direction. In circular motion the direction changes continuously.

2 Angles, radians, and angular velocity

Before we introduce any new symbol, we should pause and ask why the usual Cartesian description is not always the most economical one. If a particle is constrained to a fixed circle, then its distance from the centre is already known: it is always R. The quantity that really changes is not the distance from the centre, but the direction of the radius. The angle is therefore not an artificial complication. It is the variable that follows the geometry of the problem.

This is a good example of mathematical modelling. We do not choose variables because they look familiar. We choose variables because they make the physical situation readable. For a projectile, horizontal and vertical coordinates were natural because gravity separated the motion into two directions. For a point moving around a circle, the natural variable is the angle swept out by the radius.

Circular motion is naturally described using an angle. If a point moves along a circle of radius R, we can describe its position by the angle $\theta(t)$ measured from the positive x-axis.



The next relation is simple, but it should not be rushed. The length of the path along the circle is not the same thing as the radius. If the particle moves through a small angle, it travels a small arc. If the same angle is swept on a larger circle, the arc is longer. This is why the radius and the angle must appear together.

If s is the arc length travelled along the circle, then

$$s = R\theta,$$

provided that θ is measured in radians. This is the reason radians are natural in circular motion. Radians are not just another unit for angles; they are the angle measure for which the relation between angle and arc length is as simple as possible.

2.1 Angular velocity

Now we translate the phrase "how fast the direction changes" into calculus. In straight-line motion we measured how quickly position changes. Here the changing quantity is the angle. Therefore the analogue of ordinary velocity is not yet v, but the rate of change of θ.

Angular velocity is the rate of change of angle:

$$\omega = \frac{\mathrm{d}\theta}{\mathrm{d}t}.$$

If the angular velocity is constant, then the angle increases by equal amounts in equal times. This is the rotational analogue of uniform motion on a line, where $x(t) = x_0 + vt$. The formula below is not a new idea; it is the old idea of constant rate, written for an angle rather than for a position.

If the angular velocity is constant, then

$$\theta(t) = \theta_0 + \omega t.$$

The word *period* introduces a different way of asking the same question. Instead of asking for radians per second, we ask: how much time is needed to return to the same direction? One complete turn corresponds to an angle 2π, so the period and angular velocity must be tied together.

The period T is the time needed for one complete revolution. One revolution corresponds to an angular change of 2π. Therefore,

$$\omega T = 2\pi,$$

so

$$\boxed{\omega = \frac{2\pi}{T}}.$$

The frequency f is the number of revolutions per unit time:

$$f = \frac{1}{T}.$$

Thus

$$\omega = 2\pi f.$$

Radians are not optional in calculus

When we differentiate expressions such as $\sin\theta$ or $\cos\theta$ using the usual rules, θ must be measured in radians. Degrees are useful for description, but radians are the natural language of calculus.

2.2 Linear speed and angular speed

At this point a common student mistake appears. Students often treat v and ω as almost the same quantity. They are not the same. Angular speed tells us how quickly the radius turns. Linear speed tells us how quickly the particle moves along the path. A small wheel and a large wheel may have the same angular speed, but a point on the larger wheel travels a longer distance in the same time.

To connect these two descriptions, we go back to the arc length relation. This is exactly the same logic as in projectile motion: first identify the geometric relation, then differentiate or solve according to the question.

The arc length is

$$s = R\theta.$$

Differentiate with respect to time:

$$\frac{\mathrm{d}s}{\mathrm{d}t} = R\frac{\mathrm{d}\theta}{\mathrm{d}t}.$$

Therefore

$$v = R\omega.$$

This is not a mysterious new law. It is just the derivative of arc length. The important point is the direction of reasoning: we did not memorize $v = R\omega$ as an isolated formula. We started from the geometry of a circle and asked how distance along the curve changes when the angle changes.

Meaning of $v = R\omega$

For the same angular velocity, a point farther from the centre travels a larger distance in the same time. That is why the outer edge of a rotating disk has a larger linear speed than a point near the centre.

3 Parametric description of uniform circular motion

We now want a complete mathematical description of the position. The angle tells us where the radius points, but the particle still lives in the plane. To connect the angular description with ordinary coordinates, we project the radius onto the x- and y-axes. This is where sine and cosine enter the story. They are not decorative functions inserted into a formula; they are the coordinate shadows of a rotating radius.

Suppose a particle moves on a circle of radius R with constant angular velocity ω. A standard parametrisation is

$$x(t) = R\cos(\omega t + \varphi),$$
$$y(t) = R\sin(\omega t + \varphi),$$

where φ is the initial phase. The position vector is

$$\mathbf{r}(t) = R\cos(\omega t + \varphi)\mathbf{i} + R\sin(\omega t + \varphi)\mathbf{j}.$$

Before differentiating, we should read what the equations say. At each instant, the point has two coordinates. Each coordinate oscillates between $-R$ and R, but the two oscillations are shifted relative to each other. Together they keep the distance from the origin constant. This is a perfect example of why looking at a formula without interpretation is dangerous: each coordinate looks like one-dimensional oscillation, but the ordered pair describes motion around a circle.

This pair of equations should be read carefully. It does not say that the particle moves sinusoidally in space along a line. It says that each coordinate is sinusoidal while the pair of coordinates together traces a circle.

3.1 Velocity

The next step is not an algebraic ritual. Velocity is defined as the derivative of position, so if we want to know the velocity vector, differentiation is the mathematical translation of the physical question. We are not differentiating because the page asks us to. We are differentiating because the word "velocity" means rate of change of position.

Differentiate the position vector:

$$\mathbf{v}(t) = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}.$$

Since

$$\frac{\mathrm{d}}{\mathrm{d}t}\cos(\omega t + \varphi) = -\omega\sin(\omega t + \varphi),$$
$$\frac{\mathrm{d}}{\mathrm{d}t}\sin(\omega t + \varphi) = \omega\cos(\omega t + \varphi),$$

we obtain

$$\mathbf{v}(t) = -R\omega\sin(\omega t + \varphi)\mathbf{i} + R\omega\cos(\omega t + \varphi)\mathbf{j}.$$

Its magnitude is

$$|\mathbf{v}(t)| = \sqrt{R^2\omega^2\sin^2(\omega t + \varphi) + R^2\omega^2\cos^2(\omega t + \varphi)}.$$

Using $\sin^2 u + \cos^2 u = 1$,

$$|\mathbf{v}(t)| = R\omega.$$

Thus the speed is constant.

This conclusion deserves a moment of attention. The velocity vector is changing because its components are changing, but its magnitude is not changing. That is the central tension of uniform circular motion: the motion is uniform in speed, but not uniform in velocity. The particle is not accelerating because it goes faster and slower; it accelerates because the direction of the velocity is continually being turned.

3.2 Perpendicularity of position and velocity

Now we verify algebraically what the geometry already suggested. If the particle remains on the circle, its instantaneous motion should be tangent to the circle. Tangency means perpendicularity to the radius. In vector language, perpendicularity is tested by the dot product.

Compute the dot product:

$$\mathbf{r}(t) \cdot \mathbf{v}(t) = R\cos u(-R\omega \sin u) + R\sin u(R\omega \cos u),$$

where $u = \omega t + \varphi$. Therefore

$$\mathbf{r}(t) \cdot \mathbf{v}(t) = -R^2\omega \sin u \cos u + R^2\omega \sin u \cos u = 0.$$

Again, the velocity is tangent to the circle. Notice what has happened: the same fact has now appeared in three languages. Geometrically, motion along a circle is tangent. Algebraically, $\mathbf{r}\cdot\mathbf{v} = 0$. Physically, the velocity has no radial component, so the particle does not move inward or outward from the circle.

3.3 Acceleration

Acceleration is where circular motion becomes conceptually important. A student may expect zero acceleration because the speed is constant. The calculation below shows why that expectation is wrong. Acceleration is the derivative of the whole velocity vector, not only the derivative of its magnitude.

Differentiate the velocity:

$$\mathbf{a}(t) = \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}.$$

We get

$$\mathbf{a}(t) = -R\omega^2 \cos(\omega t + \varphi)\mathbf{i} - R\omega^2 \sin(\omega t + \varphi)\mathbf{j}.$$

Factor out $-\omega^2$:

$$\mathbf{a}(t) = -\omega^2 \left(R\cos(\omega t + \varphi)\mathbf{i} + R\sin(\omega t + \varphi)\mathbf{j}\right).$$

But the expression in parentheses is exactly $\mathbf{r}(t)$. Hence

$$\boxed{\mathbf{a}(t) = -\omega^2\mathbf{r}(t)}.$$

This compact vector equation is the moment where the model becomes powerful. It says more than the magnitude of the acceleration. It gives the direction automatically. Since $-\mathbf{r}(t)$ points from the particle back toward the centre, the equation tells us that the acceleration is inward at every instant.

This is one of the most important conclusions in the whole document. The acceleration points opposite to the position vector, therefore toward the centre of the circle.

Since $|\mathbf{r}| = R$, the magnitude is

$$a = \omega^2 R.$$

Using $v = R\omega$, we can also write

$$a = \frac{v^2}{R}.$$

These two versions answer different modelling needs. If the rotation rate is known, $a = \omega^2 R$ is natural. If the linear speed along the path is known, $a = v^2/R$ is natural. The physics is the same; the form of the equation adapts to the data and to the question.

Centripetal acceleration

Uniform circular motion has constant speed but nonzero acceleration. The acceleration points toward the centre and has magnitude

$$\boxed{a_c = \omega^2 R = \frac{v^2}{R}}.$$

4 What does "centripetal" mean?

This section is included because it prevents one of the most persistent mistakes in elementary mechanics. After deriving $a_c = v^2/R$, it is tempting to invent a new force called "the centripetal force". That is not the right interpretation. The calculation tells us what the net force must accomplish. It does not tell us which real interaction is responsible.

The word *centripetal* means centre-seeking. It describes the direction of the acceleration or of the net force required for circular motion. It is not the name of a new force in nature.

This point causes many errors. Students sometimes say, "there is gravity and centripetal force." In an orbit, this is usually the wrong language. Gravity is the force. It acts toward the centre and therefore plays the role of the centripetal force.

The correct sentence is:

The net inward force equals mv^2/R.

This sentence is deliberately phrased as a role. In one problem the role is played by tension. In another problem it is played by friction. In an orbit it is played by gravity. In a magnetic field it is played by the Lorentz force. The same mathematical requirement appears, but the physical source of the force changes.

Depending on the physical system, the inward force may be supplied by:

- tension in a string,
- static friction between tyres and road,
- gravity,
- the electric Coulomb force,
- the magnetic Lorentz force,
- the normal force or a combination of forces.

Do not add a fake extra force

In an inertial frame, do not add "centripetal force" as an additional physical force. Instead, identify real forces and set their radial resultant equal to mv^2/R.

4.1 Newton's second law in the radial direction

We now turn the kinematic result into dynamics. Kinematics told us what acceleration is required for circular motion. Newton's second law tells us what force is required for that acceleration. The calculation therefore has two layers: first geometry determines the acceleration, then dynamics identifies the force that can produce it.

Newton's second law says

$$\sum \mathbf{F} = m\mathbf{a}.$$

For uniform circular motion, the acceleration is radial inward:

$$\mathbf{a} = -\omega^2 \mathbf{r}.$$

Its magnitude is

$$a_c = \frac{v^2}{R} = \omega^2 R.$$

Therefore the radial component of the net force must satisfy

$$\boxed{\sum F_{\text{inward}} = m\frac{v^2}{R} = m\omega^2 R}.$$

The direction is part of the statement. This is why a purely scalar use of the formula can be misleading. The magnitude mv^2/R is only half of the information. The other half is that the relevant component of the net force must be inward, toward the centre of the circular path. The equation is not just about magnitudes; it says that the net force must point inward if the circular motion is uniform.

4.2 The standard solution pattern

The purpose of a pattern is not to replace thinking. The purpose is to organise thinking. In circular-motion problems, the most important first step is almost always to locate the centre and the radius of the actual path. Only then does it make sense to talk about inward direction and radial force.

Many circular-motion problems follow the same pattern:

Circular-motion problem pattern

1. Draw the circle and choose the centre.
2. Identify the radius R of the actual circular path.
3. Mark the inward radial direction.
4. List the real forces acting on the body.
5. Take the radial component of Newton's second law:

$$\sum F_{\text{inward}} = m\frac{v^2}{R}.$$

6. If needed, use $v = \omega R$ or $\omega = 2\pi/T$.
7. Interpret the result physically: what changes when R, v, m, or the force parameter changes?

This is the circular-motion equivalent of the projectile-motion pattern: situation, model, condition, mathematics, interpretation. The student should learn to say: "I am not searching for a formula. I am identifying the real force that must supply the inward acceleration required by the geometry."

5 A second derivation using changing direction

The previous derivation used coordinates and calculus. That derivation is powerful, but it can hide the physical reason for the acceleration. We now repeat the argument in a more geometric way, because students often understand circular motion better when they see that the velocity vector itself rotates.

The derivative calculation is clean, but it can hide the geometric meaning. Let us derive the centripetal acceleration again using the fact that the velocity vector changes direction.

Consider a particle moving with constant speed v along a circle of radius R. In a short time interval Δt, it travels a small arc length

$$\Delta s \approx v\Delta t.$$

The corresponding angular change is

$$\Delta\theta \approx \frac{\Delta s}{R} = \frac{v\Delta t}{R}.$$

The magnitude of the velocity does not change, but its direction turns by approximately $\Delta\theta$. The change in velocity vector has magnitude approximately

$$|\Delta\mathbf{v}| \approx v\Delta\theta.$$

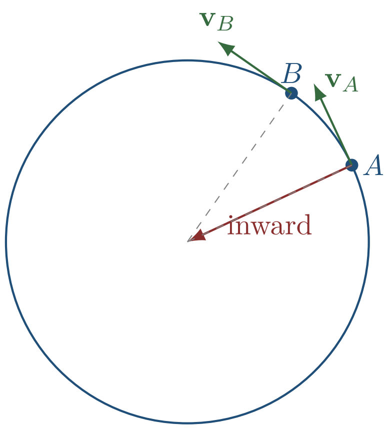
Substitute the expression for $\Delta\theta$:

$$|\Delta\mathbf{v}| \approx v\frac{v\Delta t}{R} = \frac{v^2}{R}\Delta t.$$

Therefore

$$|\mathbf{a}| = \lim_{\Delta t\to 0}\frac{|\Delta\mathbf{v}|}{\Delta t} = \frac{v^2}{R}.$$

The direction of $\Delta\mathbf{v}$ tends toward the centre as Δt becomes small. Hence the acceleration is inward.



This derivation is less algebraic but more conceptual. It shows why acceleration appears even when the speed is constant: the velocity vector rotates.

6 Uniform versus non-uniform circular motion

So far we have focused on the cleanest case: constant speed around a circle. Real circular motion may be more complicated. A car entering a roundabout may speed up or slow down. A bead moving around a vertical hoop changes speed under gravity. In those cases the path may still be circular, but the acceleration has more than one job.

Uniform circular motion means constant speed. But a particle can move on a circle with changing speed. Then the acceleration has two components:

- a radial component, responsible for changing the direction of velocity;
- a tangential component, responsible for changing the magnitude of velocity.

Let the angular position be $\theta(t)$. Define

$$\omega = \frac{\mathrm{d}\theta}{\mathrm{d}t}, \qquad \alpha = \frac{\mathrm{d}\omega}{\mathrm{d}t} = \frac{\mathrm{d}^2\theta}{\mathrm{d}t^2}.$$

The linear speed is

$$v = R\omega.$$

The tangential acceleration is

$$a_t = \frac{\mathrm{d}v}{\mathrm{d}t} = R\frac{\mathrm{d}\omega}{\mathrm{d}t} = R\alpha.$$

The radial acceleration is still

$$a_r = R\omega^2 = \frac{v^2}{R}.$$

Therefore, in vector form,

$$\boxed{\mathbf{a} = -R\omega^2\mathbf{e}_r + R\alpha\mathbf{e}_\theta}.$$

Here $\mathbf{e}_r$ points outward from the centre and $\mathbf{e}_\theta$ points tangentially in the direction of increasing θ. The radial term has a minus sign because centripetal acceleration points inward.

Two jobs of acceleration

Radial acceleration changes the direction of velocity. Tangential acceleration changes the speed. In uniform circular motion, the tangential component is zero, but the radial component is not.

6.1 Why this matters

This distinction is not a technical detail. It answers two different physical questions. The radial component explains why the path bends. The tangential component explains why the speed changes. When students mix these two, they often put the right numbers into the wrong physical place.

If a car speeds up while turning, the passenger feels both effects:

- the inward acceleration needed to follow the curve;
- the forward or backward tangential acceleration associated with changing speed.

The total acceleration magnitude is

$$a = \sqrt{a_r^2 + a_t^2} = \sqrt{(R\omega^2)^2 + (R\alpha)^2}.$$

But the two components have different physical meanings. This is why blindly using one formula can be dangerous: the question determines which component is relevant.

7 The radial basis: why the unit vectors change

There is another way to describe the same motion that is closer to the geometry of the circle. Instead of using fixed horizontal and vertical unit vectors, we can use a radial unit vector and a tangential unit vector. This language is elegant, but it has a trap: these unit vectors move with the particle. Since their directions change, their derivatives are not zero.

A common hidden difficulty in circular motion is that the natural directions themselves change with time. In Cartesian coordinates, $\mathbf{i}$ and $\mathbf{j}$ are fixed. In polar coordinates, $\mathbf{e}_r$ and $\mathbf{e}_\theta$ rotate with the particle.

Define

$$\mathbf{e}_r = (\cos\theta)\mathbf{i} + (\sin\theta)\mathbf{j},$$

$$\mathbf{e}_\theta = -(\sin\theta)\mathbf{i} + (\cos\theta)\mathbf{j}.$$

Then

$$\frac{\mathrm{d}\mathbf{e}_r}{\mathrm{d}\theta} = \mathbf{e}_\theta,$$

$$\frac{\mathrm{d}\mathbf{e}_\theta}{\mathrm{d}\theta} = -\mathbf{e}_r.$$

Using the chain rule,

$$\frac{\mathrm{d}\mathbf{e}_r}{\mathrm{d}t} = \frac{\mathrm{d}\mathbf{e}_r}{\mathrm{d}\theta}\frac{\mathrm{d}\theta}{\mathrm{d}t} = \omega\mathbf{e}_\theta,$$

$$\frac{\mathrm{d}\mathbf{e}_\theta}{\mathrm{d}t} = \frac{\mathrm{d}\mathbf{e}_\theta}{\mathrm{d}\theta}\frac{\mathrm{d}\theta}{\mathrm{d}t} = -\omega\mathbf{e}_r.$$

For circular motion with fixed radius R,

$$\mathbf{r} = R\mathbf{e}_r.$$

Differentiate:

$$\mathbf{v} = R\frac{\mathrm{d}\mathbf{e}_r}{\mathrm{d}t} = R\omega\mathbf{e}_\theta.$$

Differentiate again:

$$\mathbf{a} = R\frac{\mathrm{d}\omega}{\mathrm{d}t}\mathbf{e}_\theta + R\omega\frac{\mathrm{d}\mathbf{e}_\theta}{\mathrm{d}t}.$$

Therefore

$$\mathbf{a} = R\alpha\mathbf{e}_\theta - R\omega^2\mathbf{e}_r.$$

This derivation explains why a radial acceleration appears even when ω is constant. The term comes from the derivative of the tangential unit vector, not from a change in the magnitude of speed.

A common error

It is wrong to say: "If v is constant, then acceleration is zero." Acceleration is the derivative of the velocity vector, not only the derivative of the speed.

8 How a circular-motion calculation should be read

Before moving into examples, it is useful to describe the mental route that a good solution follows. A weak solution usually begins with a formula. A strong solution begins with a sentence about the situation. For example, it does not begin with

$$F = \frac{mv^2}{R}.$$

It begins with a statement such as:

> *The object is constrained to move along a circle of radius R, so its acceleration must have an inward radial component of magnitude v^2/R.*

Only after this sentence does Newton's second law enter. The equation is then not a magic tool; it is the mathematical encoding of the sentence.

A good circular-motion solution normally answers four questions before doing algebra.

Four questions before algebra

1. What object is moving?
2. What is the radius of its actual circular path?
3. Where is the centre, and which direction is inward at the instant considered?
4. Which real force or component of a force points inward and can provide the required acceleration?

These questions prevent most mistakes. For example, in a car-turning problem the inward force is not a mysterious circular-motion force. It is static friction. In a satellite problem the inward force is gravity. In a magnetic field problem the inward force is the Lorentz force. The mathematical role is the same, but the physical source is different.

The calculation then has the following form:

$$\text{available inward force} = \text{required inward resultant}.$$

For uniform circular motion, the required inward resultant is

$$m\frac{v^2}{R} = m\omega^2 R.$$

The left-hand side of the modelling equation changes from problem to problem; the right-hand side is determined by the geometry of circular motion.

The recurring modelling sentence

If the path is circular, then the net inward force is not optional. It must have the magnitude required to turn the velocity vector continuously toward the centre.

This is why circular motion is such a good recap topic. It forces us to connect geometry, calculus, force diagrams, Newton's second law, and interpretation in one chain of reasoning.

9 Example 1: a stone on a string

We begin with the simplest physical example because it exposes the whole logic without unnecessary complications. The string is easy to see. It pulls inward. If the pull is removed, the circular motion immediately disappears. This makes the connection between force and path very concrete.

A stone of mass m is attached to a string and moves in a horizontal circle of radius R with speed v. Neglect air resistance. What is the tension in the string?

9.1 Physical description

The stone follows a circular path. Its acceleration points toward the centre. The string pulls the stone toward the centre. Therefore, the tension supplies the radial force.

The model is simple:

- mass: m;
- radius: R;
- speed: v;
- inward force: tension T;
- radial acceleration: v^2/R.

9.2 Mathematical translation

The translation is short, but it carries the whole model. We are not writing $T = mv^2/R$ because it is the formula for a string. We are writing it because, in this particular situation, the tension is the only horizontal inward force and therefore the tension must equal the required centripetal resultant.

Newton's second law in the inward radial direction gives

$$T = m\frac{v^2}{R}.$$

Therefore

$$\boxed{T = \frac{mv^2}{R}}.$$

This expression is easy to remember, but the meaning is more important:

- doubling the mass doubles the required tension;
- doubling the speed quadruples the required tension;
- doubling the radius halves the required tension if speed is fixed.

9.3 What if the string breaks?

If the string breaks, the inward force disappears. The stone does not continue moving in a circle. At the instant of breaking, it moves along the tangent with its current velocity.

This is another way to see that circular motion requires a continuous inward force. Without that force, the natural inertial motion is straight-line motion.

Question behind the calculation

The calculation does not answer "which formula?" It answers: what real force must provide the inward acceleration required by the circular path?

10 Example 2: a passenger on a carousel

The carousel example is useful because it connects classroom mechanics with bodily experience. A passenger feels as if something pushes them outward. But from the ground frame, the important question is different: what inward force bends the passenger's motion into a circle? The answer is usually static friction.

A passenger stands on a rotating carousel at distance R from the centre. The carousel rotates with angular speed ω. What horizontal force is needed to keep the passenger moving in a circle?

10.1 Model

The passenger has speed

$$v = \omega R.$$

The required centripetal acceleration is

$$a_c = \omega^2 R.$$

The horizontal inward force must have magnitude

$$F = m\omega^2 R.$$

What supplies this force? Usually static friction between the passenger's shoes and the carousel floor.

10.2 Maximum angular speed before slipping

Now the problem becomes a condition problem rather than a direct calculation. Circular motion requires a certain inward force. Static friction can provide only up to a maximum value. The question "when does slipping begin?" becomes the mathematical inequality "required force is less than or equal to available force."

Let μ_s be the coefficient of static friction. The maximum static friction is

$$f_{s,\max} = \mu_s N.$$

On a horizontal carousel, $N = mg$, so

$$f_{s,\max} = \mu_s mg.$$

For circular motion without slipping, friction must be able to provide

$$m\omega^2 R \le \mu_s mg.$$

Cancel m:

$$\omega^2 R \le \mu_s g.$$

Therefore

$$\boxed{\omega \le \sqrt{\frac{\mu_s g}{R}}}.$$

10.3 Interpretation

The larger the radius, the smaller the maximum allowed angular speed. This matches experience: people near the edge of a carousel are more likely to slip outward relative to the rotating platform.

Notice that the mass cancels. A heavier passenger needs more centripetal force, but also has proportionally larger maximum friction because $N = mg$. In this simplified model, the slipping condition is independent of mass.

11 Example 3: a car turning on a flat road

The car-on-a-turn problem is the same structure as the carousel, but now the language is more familiar: tyres, road, grip, safe speed. This example is important because it shows that circular motion is not only about objects literally going around perfect circles. Any curved path has local curvature, and a turn can be approximated by a circular arc with some radius R.

A car of mass m turns on a flat road along a circular arc of radius R. What is the maximum speed before the tyres lose static grip?

11.1 Forces

A good solution begins by separating directions. Vertically, the car is not lifting off the road and not sinking through it, so the vertical acceleration is zero. Horizontally, the car is changing direction, so there must be an inward acceleration. This is why the same force diagram must be read differently in the two directions.

The vertical forces are:

$$N \quad \text{upward}, \qquad mg \quad \text{downward}.$$

There is no vertical acceleration, so

$$N = mg.$$

The horizontal inward force is static friction. Its maximum value is

$$f_{s,\max} = \mu_s N = \mu_s mg.$$

For circular motion, the required inward force is

$$m\frac{v^2}{R}.$$

Therefore the condition for not slipping is

$$m\frac{v^2}{R} \le \mu_s mg.$$

Cancel m:

$$\frac{v^2}{R} \le \mu_s g.$$

Thus

$$\boxed{v \le \sqrt{\mu_s g R}}.$$

11.2 Physical reading

The maximum safe speed increases with the square root of the radius. A gentle curve can be taken faster than a sharp curve. The maximum speed also increases with friction. Wet or icy roads have smaller μ_s, so the maximum safe speed decreases.

Static friction is not automatically equal to $\mu_s N$

Static friction adjusts up to a maximum. We use $f_s = \mu_s N$ only at the limiting case of impending slip.

11.3 The false outward force

In the ground frame, the car accelerates inward. The force causing this acceleration is static friction directed inward. There is no real outward force acting on the car in the inertial ground frame.

Passengers may feel pushed outward because their bodies tend to continue in a straight line while the car turns inward. In a rotating frame one may introduce a fictitious centrifugal force, but in the inertial frame the real force is inward.

12 Banked curves

A banked curve is a beautiful example of engineering encoded in mechanics. Instead of asking friction to do all the horizontal work, the road is tilted so that the normal force itself has an inward component. The road geometry is designed so that the force geometry matches the required acceleration.

A banked road is tilted so that the normal force has an inward horizontal component. This allows a car to turn even with less reliance on friction.

Consider a frictionless banked curve of radius R and banking angle β. The forces are:

- weight mg downward;
- normal force N perpendicular to the road surface.

The calculation below is not a trick with trigonometry. It is the mathematical translation of a force diagram. The same normal force must do two things at once: its vertical component balances weight, and its horizontal component supplies the inward acceleration.

Resolve N into vertical and horizontal components:

$$N \cos\beta = mg,$$

$$N \sin\beta = m\frac{v^2}{R}.$$

Divide the second equation by the first:

$$\tan\beta = \frac{v^2}{Rg}.$$

Thus

$$\boxed{v = \sqrt{Rg\tan\beta}}.$$

12.1 Interpretation

This is the design speed for a frictionless banked curve. At this speed, the normal force alone supplies the necessary inward component. If the car moves much faster or slower, friction becomes necessary to prevent slipping.

Again, the equation is not a memorized object. It comes from asking which component of which real force supplies the centripetal acceleration.

13 Gravity as the centripetal force

The power of the circular-motion model becomes much clearer when we move to astronomy. Nothing physical is tied to a string, and there is no road or friction. Yet the same structure appears. The orbit is circular, so the acceleration must be inward. Gravity already points inward, so gravity is the real force that plays the centripetal role.

Now we move from everyday examples to orbital motion. Suppose a body of mass m moves in a circular orbit of radius r around a much larger mass M. The gravitational force has magnitude

$$F_g = G\frac{Mm}{r^2}.$$

This force law contains two different pieces of information. Its magnitude is GMm/r^2, and its direction is toward the central mass. The direction is exactly what circular motion needs. Therefore, for a circular orbit, the modelling step is to equate the gravitational force with the required inward resultant.

It points toward the centre of the larger mass. If the orbit is circular, this gravitational force supplies the required centripetal force:

$$G\frac{Mm}{r^2} = m\frac{v^2}{r}.$$

Cancel m:

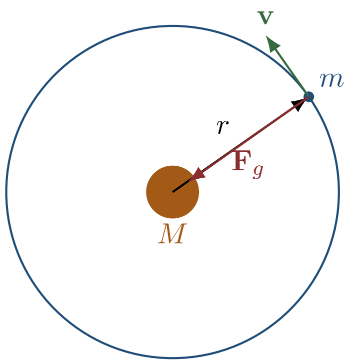
$$G\frac{M}{r^2} = \frac{v^2}{r}.$$

Multiply by r:

$$v^2 = \frac{GM}{r}.$$

Therefore

$$\boxed{v = \sqrt{\frac{GM}{r}}}.$$



13.1 Physical interpretation

The algebra has produced a simple expression, but the expression is not the end of the reasoning. We must read it. Reading an equation means asking what changes when one parameter changes and what quantities do not appear at all.

The orbital speed decreases with radius:

$$v \propto \frac{1}{\sqrt{r}}.$$

A satellite in a higher circular orbit moves more slowly than one in a lower circular orbit. This can feel counterintuitive, but it follows directly from balancing gravity with the centripetal requirement.

The orbiting mass m cancels. In this simplified two-body circular model, the orbital speed does not depend on the satellite's mass.

13.2 Period of a circular orbit

Speed tells us how fast the satellite moves along the orbit, but often the observable quantity is the period: the time for one complete revolution. To move from speed to period, we return to geometry. One circular orbit has length $2\pi r$, and time equals distance divided by speed.

The period is circumference divided by speed:

$$T = \frac{2\pi r}{v}.$$

Substitute

$$v = \sqrt{\frac{GM}{r}}.$$

Then

$$T = \frac{2\pi r}{\sqrt{GM/r}}.$$

This is

$$T = 2\pi\sqrt{\frac{r^3}{GM}}.$$

Squaring gives

$$\boxed{T^2 = \frac{4\pi^2}{GM}r^3}.$$

This is the circular-orbit form of Kepler's third law:

$$T^2 \propto r^3.$$

Kepler's third law from circular motion

For circular orbits around the same central mass,

$$\frac{T^2}{r^3} = \frac{4\pi^2}{GM}.$$

The proportionality constant contains information about the central mass.

14 Reading Kepler's third law as a model

This section is deliberately interpretive. Kepler's third law is often memorized as a proportion, but in this course we want to see it as the visible trace of a model. The exponent 3 in $T^2 \propto r^3$ is not arbitrary. It comes from combining circular kinematics with an inverse-square force.

The equation

$$T^2 = \frac{4\pi^2}{GM} r^3$$

is often treated as a formula. But it is better understood as a compressed conclusion from several assumptions:

1. the orbit is circular;
2. the central mass is much larger than the orbiting mass;
3. gravity is Newtonian and follows the inverse-square law;
4. the gravitational force supplies the centripetal force;
5. the period is circumference divided by speed.

Each assumption matters. If the force law were different, the relation between T and r would change. If the orbit were highly elliptical, the circular derivation would not be literally correct, although Kepler's third law has a more general elliptical version using the semi-major axis.

14.1 How the inverse-square law leaves a fingerprint

Here we step back and ask a more mathematical question. What part of Kepler's law is specific to gravity? If the central force depended on distance in a different way, the period-radius relation would change. This is a powerful modelling idea: observed motion can reveal the force law that produces it.

Suppose a central force has the general form

$$F(r) = \frac{K}{r^n}.$$

For a circular orbit,

$$\frac{K}{r^n} = m\frac{v^2}{r}.$$

Thus

$$v^2 \propto r^{1-n}.$$

Since

$$T = \frac{2\pi r}{v},$$

we get

$$T^2 \propto \frac{r^2}{v^2} \propto \frac{r^2}{r^{1-n}} = r^{n+1}.$$

For Newtonian gravity, $n = 2$, so

$$T^2 \propto r^3.$$

This is a powerful interpretation. The exponent in the orbital period relation is connected to the exponent in the force law.

What is being asked?

If a problem asks for the period-radius relation, the mathematical task is not merely to substitute values. The task is to combine a force law with the centripetal condition and then interpret the resulting dependence.

15 Energy in a circular gravitational orbit

The force analysis tells us the speed required for a circular orbit. Energy gives a different reading of the same orbit. It answers not "what force points inward right now?" but "how are kinetic and potential energy related in this bound state?" The two perspectives should agree, but they emphasize different aspects of the model.

For a circular gravitational orbit,

$$G\frac{Mm}{r^2} = m\frac{v^2}{r}.$$

We found

$$v^2 = \frac{GM}{r}.$$

The kinetic energy is

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\frac{GM}{r} = \frac{GMm}{2r}.$$

The gravitational potential energy, with zero at infinity, is

$$U = -\frac{GMm}{r}.$$

Therefore the total energy is

$$E = K + U = \frac{GMm}{2r} - \frac{GMm}{r} = -\frac{GMm}{2r}.$$

So

$$\boxed{E = -\frac{GMm}{2r}}.$$

This negative total energy means the orbit is bound. The satellite does not have enough energy to escape to infinity with zero speed.

15.1 Why this belongs in circular motion

The energy result uses the circular-orbit condition. Without the relation $v^2 = GM/r$, we could not reduce the kinetic energy to a function of r alone. Again, the circular model converts dynamics into a relation among quantities.

16 Coulomb attraction and circular motion

The next example shows how abstract the circular-motion structure really is. Gravity is not the only inverse-square attraction. Electric charges also exert forces that can point toward or away from each other. If the charges have opposite signs, the Coulomb force is attractive and can play the same centripetal role as gravity.

The gravitational force between masses and the electric force between charges have similar mathematical forms. For two opposite charges of magnitudes q_1 and q_2, the Coulomb attraction has magnitude

$$F_C = k\frac{|q_1 q_2|}{r^2},$$

where

$$k = \frac{1}{4\pi\varepsilon_0}.$$

Suppose a light particle of mass m moves in a circular orbit around a much heavier oppositely charged centre. The electric force supplies the centripetal force:

$$k\frac{|q_1 q_2|}{r^2} = m\frac{v^2}{r}.$$

Thus

$$v^2 = \frac{k|q_1 q_2|}{mr}.$$

So

$$\boxed{v = \sqrt{\frac{k|q_1 q_2|}{mr}}}.$$

16.1 Comparison with gravity

This comparison is more important than the formulas themselves. The symbols change, but the modelling sentence is the same: an inverse-square central attraction supplies the inward force needed for a circular path. The analogy helps students see physics as structure rather than as disconnected chapters.

For gravity:

$$v^2 = \frac{GM}{r}.$$

For Coulomb attraction:

$$v^2 = \frac{k|q_1 q_2|}{mr}.$$

The mathematical shape is the same: $v^2 \propto 1/r$. The physical constants are different.

This is a central point of the document. Once we understand circular motion, a new force law can be inserted into the same structural equation:

$$\text{inward force} = m\frac{v^2}{r}.$$

16.2 Period

The period is

$$T = \frac{2\pi r}{v}.$$

Substituting the expression for v:

$$T = 2\pi r\sqrt{\frac{mr}{k|q_1 q_2|}}.$$

Therefore

$$\boxed{T = 2\pi\sqrt{\frac{mr^3}{k|q_1 q_2|}}}.$$

Again,

$$T^2 \propto r^3,$$

because Coulomb attraction is also an inverse-square force.

17 The classical electron model and its limitation

At this point it is tempting to say: if Coulomb attraction can provide centripetal acceleration, then perhaps an atom is just a tiny planetary system. Historically, this was a natural thought. But it is not a complete physical theory. The point of this section is to use circular motion as a bridge toward the Bohr model, while also making clear why the purely classical picture fails.

One might imagine an electron moving in a circular orbit around a proton, held by the Coulomb attraction. Classically, the circular-motion condition would be

$$k\frac{e^2}{r^2} = m_e\frac{v^2}{r}.$$

This gives

$$v^2 = \frac{ke^2}{m_e r}.$$

But classical electrodynamics says that an accelerating charge radiates energy. A charge in circular motion is accelerating, so a purely classical electron orbit would lose energy and spiral inward. This is one reason why classical physics is not enough to explain atomic stability.

The Bohr model adds a new quantization condition. It is not the final quantum theory, but it is historically and pedagogically useful because it combines circular motion with a quantum postulate.

18 The Bohr model as circular motion plus quantization

The Bohr model is not introduced here as modern atomic theory. It is introduced because it is a historically important example of how a familiar mechanical structure can be modified by a new physical principle. The circular-motion equation supplies one relation. Quantization of angular momentum supplies another. Together they restrict the allowed radii.

In the simplest Bohr model of hydrogen, an electron moves in a circular orbit around a proton. The Coulomb force supplies the centripetal force:

$$k\frac{e^2}{r^2} = m_e\frac{v^2}{r}.$$

This is the classical circular-motion part.

Bohr adds the angular momentum quantization condition:

$$m_e vr = n\hbar,$$

where

$$n = 1, 2, 3, \dots$$

and

$$\hbar = \frac{h}{2\pi}.$$

The model now has two equations:

$$k\frac{e^2}{r^2} = m_e\frac{v^2}{r},$$
$$m_e vr = n\hbar.$$

The unknowns are v and r for each integer n.

18.1 Solving for allowed radii

Now we solve the model slowly. The important feature is that we have two equations for the two unknowns r and v. The first equation is dynamical: Coulomb attraction supplies centripetal force. The second is quantum: angular momentum is allowed only in multiples of $\hbar$. The allowed radii come from combining these two statements.

From angular momentum quantization,

$$v = \frac{n\hbar}{m_e r}.$$

Substitute into the circular-motion equation:

$$k\frac{e^2}{r^2} = m_e\frac{1}{r}\left(\frac{n\hbar}{m_e r}\right)^2.$$

Simplify the right-hand side:

$$m_e\frac{1}{r}\frac{n^2\hbar^2}{m_e^2 r^2} = \frac{n^2\hbar^2}{m_e r^3}.$$

Thus

$$k\frac{e^2}{r^2} = \frac{n^2\hbar^2}{m_e r^3}.$$

Multiply by r^3:

$$ke^2 r = \frac{n^2\hbar^2}{m_e}.$$

Therefore

$$\boxed{r_n = \frac{n^2\hbar^2}{m_e k e^2}}.$$

The radius grows like

$$r_n \propto n^2.$$

The ground-state radius is

$$a_0 = \frac{\hbar^2}{m_e k e^2},$$

called the Bohr radius. Hence

$$\boxed{r_n = n^2 a_0}.$$

18.2 Allowed speeds

Using

$$m_e v r = n\hbar,$$

we have

$$v_n = \frac{n\hbar}{m_e r_n}.$$

Substitute $r_n = n^2 a_0$:

$$v_n = \frac{n\hbar}{m_e n^2 a_0} = \frac{\hbar}{m_e a_0}\frac{1}{n}.$$

Using the expression for a_0,

$$\frac{\hbar}{m_e a_0} = \frac{ke^2}{\hbar}.$$

Therefore

$$\boxed{v_n = \frac{ke^2}{\hbar}\frac{1}{n}}.$$

The allowed speed decreases as n increases.

19 Energy levels in the Bohr model

Once the allowed radii are known, the next question is energy. Energy is the quantity that connects the mechanical orbit to spectral lines. Again, we are not merely substituting into a memorized expression; we are using the radius and speed restrictions to see which energies the model permits.

The kinetic energy is

$$K = \frac{1}{2}m_e v^2.$$

From the circular-motion equation,

$$m_e v^2 = \frac{ke^2}{r}.$$

Therefore

$$K = \frac{1}{2}\frac{ke^2}{r}.$$

The electric potential energy is

$$U = -\frac{ke^2}{r}.$$

Thus the total energy is

$$E = K + U = \frac{ke^2}{2r} - \frac{ke^2}{r} = -\frac{ke^2}{2r}.$$

For the allowed Bohr radius $r_n = n^2 a_0$,

$$E_n = -\frac{ke^2}{2n^2 a_0}.$$

So

$$\boxed{E_n \propto -\frac{1}{n^2}}.$$

More explicitly,

$$E_n = -\frac{m_e k^2 e^4}{2\hbar^2}\frac{1}{n^2}.$$

19.1 Interpretation

The energy levels are discrete because the angular momentum condition allows only certain radii. The classical circular-motion equation gives a continuum of possible circular orbits. Bohr's quantization rule selects a discrete subset.

What circular motion contributes to the Bohr model

Circular motion supplies the relation between Coulomb force, radius, and speed. Quantization supplies the additional rule that not every classical circular orbit is allowed.

19.2 Why this model is useful but limited

The Bohr model is not modern quantum mechanics. Electrons are not tiny planets literally moving on classical circular orbits. Still, the model is useful for teaching because it shows how a familiar mechanical structure can combine with a new physical postulate to produce discrete radii and energies.

The mathematical lesson remains valuable: a physical model is built by combining conditions. Here the conditions are:

1. inward Coulomb force provides centripetal acceleration;
2. angular momentum is quantized;
3. energy is kinetic plus electric potential energy.

20 Magnetic Lorentz force and circular motion

Magnetic circular motion is different from gravitational and electric attraction in one important way. The magnetic force is not a central force pointing toward a fixed source. Instead, it is perpendicular to the velocity. But a force perpendicular to velocity can bend the path without changing the speed. That is exactly what uniform circular motion requires.

A charged particle moving in a magnetic field experiences the magnetic part of the Lorentz force:

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B}.$$

If the velocity is perpendicular to the magnetic field, the magnitude is

$$F = |q|vB.$$

The force is perpendicular to the velocity. Since it changes the direction of velocity but not its magnitude, it can produce circular motion.

This sentence is the whole physical reason for circular magnetic motion. A magnetic force perpendicular to velocity does no work, because the displacement is along the velocity while the force is perpendicular to it. Therefore the speed can remain constant even though the direction changes. The magnetic field changes the direction of motion, not the kinetic energy.

Set magnetic force equal to centripetal force:

$$|q|vB = m\frac{v^2}{r}.$$

Assuming $v \neq 0$, cancel one factor of v:

$$|q|B = m\frac{v}{r}.$$

Therefore

$$\boxed{r = \frac{mv}{|q|B}}.$$

20.1 Angular frequency

The radius depends on speed, but the angular frequency has a surprising simplification. Once we write $v = \omega r$, the speed cancels. This cancellation has physical meaning: in a uniform magnetic field, faster particles move in larger circles, but they complete rotations with the same angular frequency if q, m, and B are fixed.

Using $v = \omega r$,

$$|q|vB = m\frac{v^2}{r} = mv\omega.$$

Cancel v:

$$|q|B = m\omega.$$

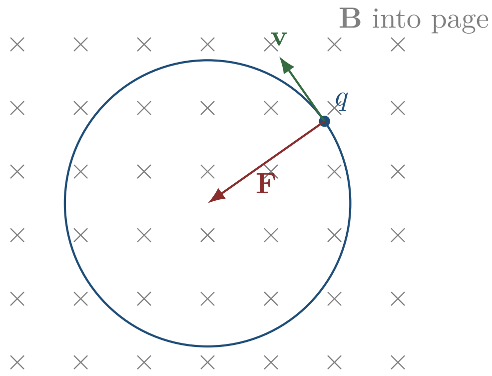
Thus

$$\boxed{\omega = \frac{|q|B}{m}}.$$

The period is

$$T = \frac{2\pi}{\omega} = \frac{2\pi m}{|q|B}.$$

A remarkable result appears: for nonrelativistic motion in a uniform magnetic field, the angular frequency does not depend on the speed. Faster particles move in larger circles but complete revolutions in the same time.



20.2 Direction of rotation

The direction depends on the sign of q and the direction of $\mathbf{B}$. The right-hand rule gives the direction of $\mathbf{v} \times \mathbf{B}$ for a positive charge. A negative charge accelerates in the opposite direction.

The magnitude equations determine the radius and frequency. The vector equation determines the direction of curvature.

21 Helical motion in a magnetic field

The purely circular case occurs only when the velocity is perpendicular to the magnetic field. If the velocity also has a component along the field, that component is unaffected by the magnetic force. The result is a superposition: circular motion around the field direction plus uniform motion along the field direction.

If the velocity has a component perpendicular to $\mathbf{B}$ and a component parallel to $\mathbf{B}$, the perpendicular component produces circular motion while the parallel component remains unchanged.

Write

$$\mathbf{v} = \mathbf{v}_\perp + \mathbf{v}_\parallel.$$

The magnetic force is perpendicular to $\mathbf{v}$ and $\mathbf{B}$. The component parallel to $\mathbf{B}$ does not contribute to the force because

$$\mathbf{v}_\parallel \times \mathbf{B} = \mathbf{0}.$$

Thus the radius of the circular part is

$$r = \frac{mv_\perp}{|q|B}.$$

The angular frequency is still

$$\omega = \frac{|q|B}{m}.$$

The motion is a helix: circular motion around the field direction combined with uniform motion along the field direction.

21.1 Pitch of the helix

The period of one circular revolution is

$$T = \frac{2\pi m}{|q|B}.$$

During this time, the particle moves parallel to the field by distance

$$p = v_\parallel T.$$

Therefore the pitch is

$$\boxed{p = v_\parallel \frac{2\pi m}{|q|B}}.$$

Again, we did not memorize a special helix formula. We decomposed the velocity into components and applied circular motion to the perpendicular part.

22 Comparing central forces

At this point the examples may look numerous, but the underlying logic is small. The table below should be read as a map of roles. In each row, the real force is different. The circular-motion requirement is the same. The art of solving the problem is to identify which physical interaction supplies the inward resultant.

The following table shows how the same circular-motion condition appears in different contexts.

System	Inward force	Circular-motion condition
Stone on string	tension T	$T = mv^2/R$
Car on flat curve	static friction f_s	$f_s = mv^2/R$
Gravitational orbit	GMm/r^2	$GMm/r^2 = mv^2/r$
Coulomb attraction	$k\|q_1q_2\|/r^2$	$k\|q_1q_2\|/r^2 = mv^2/r$
Magnetic field	$\|q\|vB$	$\|q\|vB = mv^2/r$

The left side changes. The right side is the kinematic requirement of circular motion.

23 A full modelling example: choosing the equation

This example is included to slow down the decision process. Many students want to jump immediately to an equation. Instead, we first identify the object, the path, the force that can point inward, and the condition that prevents failure of the motion. Only after that do we write the equation.

Consider the following question:

> A particle moves in a circle of radius R. Its period is T. What is the magnitude of the net force required to maintain this motion?

A weak approach is to search memory for a force formula. A better approach is to translate the information.

The period gives angular speed:

$$\omega = \frac{2\pi}{T}.$$

The speed is

$$v = \omega R = \frac{2\pi R}{T}.$$

The required acceleration is

$$a_c = \frac{v^2}{R}.$$

Substitute v:

$$a_c = \frac{1}{R}\left(\frac{2\pi R}{T}\right)^2 = \frac{4\pi^2 R}{T^2}.$$

Thus the net inward force must be

$$\boxed{F = m\frac{4\pi^2 R}{T^2}}.$$

23.1 What exactly was used?

We used three relations, each with a different meaning:

- $\omega = 2\pi/T$: definition of period and angular speed;
- $v = \omega R$: geometry of arc length;
- $F = ma_c$: Newton's second law with centripetal acceleration.

This is why the final expression is not an isolated formula. It is a compressed record of the modelling chain.

The modelling chain

Period T determines angular speed. Angular speed and radius determine linear speed. Linear speed and radius determine acceleration. Acceleration and mass determine net force.

24 Inverse problems

A formula becomes useful only when we understand that it is a relation. The same relation can be read forward or backward. Sometimes we know the speed and ask for the required force. Sometimes we know the maximum available force and ask for the maximum speed. Sometimes we know the speed and force and ask what radius is possible. These are not different laws; they are different questions asked of the same model.

Circular-motion equations can be read in many directions. This is another reason formula memorization is weak. A relation is more flexible than a recipe.

Suppose the inward force has fixed maximum magnitude $F_{\max}$. Then circular motion of radius R and speed v requires

$$m\frac{v^2}{R} \le F_{\max}.$$

This can answer different questions.

24.1 Maximum speed

Solve for v:

$$v^2 \le \frac{F_{\max}R}{m},$$

so

$$\boxed{v \le \sqrt{\frac{F_{\max}R}{m}}}.$$

24.2 Minimum radius

Solve for R:

$$R \ge \frac{mv^2}{F_{\max}}.$$

Thus

$$\boxed{R_{\min} = \frac{mv^2}{F_{\max}}}.$$

24.3 Maximum angular velocity

Using $v = \omega R$,

$$m\omega^2 R \le F_{\max}.$$

Therefore

$$\boxed{\omega \le \sqrt{\frac{F_{\max}}{mR}}}.$$

The physical situation is the same. The unknown changes. Mathematics lets us read the same relation from different angles.

25 Vertical circular motion

Vertical circular motion is a useful warning against mechanical use of formulas. The acceleration required for circular motion still points toward the centre, but gravity has different radial components at different points of the path. At the top and bottom, the equations look different because the direction of the centre relative to gravity is different.

Vertical circular motion adds an important complication: gravity has different radial components at different points of the circle.

Consider a mass attached to a string and moving in a vertical circle of radius R. At the top of the circle, both gravity and tension point toward the centre. At the bottom, tension points toward the centre but gravity points away from the centre.

25.1 At the top

At the top, the centre of the circle lies downward. Gravity also points downward. If there is a string or track force, it may also point toward the centre depending on the situation. This is why the top point is often the place where contact can be lost: gravity helps provide the inward acceleration.

At the top, inward is downward. The radial equation is

$$T_{\text{top}} + mg = m\frac{v_{\text{top}}^2}{R}.$$

Thus

$$T_{\text{top}} = m\frac{v_{\text{top}}^2}{R} - mg.$$

For the string to remain taut,

$$T_{\text{top}} \geq 0.$$

Therefore

$$m\frac{v_{\text{top}}^2}{R} - mg \geq 0,$$

so

$$v_{\text{top}}^2 \geq gR.$$

Thus the minimum speed at the top is

$$\boxed{v_{\text{top,min}} = \sqrt{gR}}.$$

25.2 At the bottom

At the bottom, the centre lies upward while gravity points downward. Therefore gravity works against the required inward direction. The normal force or tension must not only overcome weight but also provide the centripetal requirement. This is why tension or normal force is often largest at the bottom.

At the bottom, inward is upward. The radial equation is

$$T_{\text{bottom}} - mg = m\frac{v_{\text{bottom}}^2}{R}.$$

Therefore

$$T_{\text{bottom}} = m\frac{v_{\text{bottom}}^2}{R} + mg.$$

The tension at the bottom is larger than the centripetal requirement alone, because it must both overcome weight and provide inward acceleration.

25.3 Energy connection

If air resistance is neglected and the string tension does no work, mechanical energy is conserved. Between bottom and top, the height increases by $2R$. Therefore

$$\frac{1}{2}mv_{\text{bottom}}^2 = \frac{1}{2}mv_{\text{top}}^2 + mg(2R).$$

Cancel m and multiply by 2:

$$v_{\text{bottom}}^2 = v_{\text{top}}^2 + 4gR.$$

If the minimum top speed is $v_{\text{top}}^2 = gR$, then

$$v_{\text{bottom}}^2 = 5gR.$$

Thus the minimum bottom speed needed to complete the circle with a taut string is

$$\boxed{v_{\text{bottom,min}} = \sqrt{5gR}}.$$

Why vertical circular motion is richer

The radial equation changes from point to point because the direction of gravity relative to the radius changes. The circular-motion condition remains mv^2/R, but the force balance must be written separately at each position.

26 Circular motion and harmonic motion

This section shows that circular motion is not isolated from the rest of mechanics. If we watch only one coordinate of a point moving uniformly around a circle, that coordinate oscillates sinusoidally. In this sense, harmonic motion can be seen as the shadow of uniform circular motion.

Uniform circular motion is closely connected to simple harmonic motion. The projection of uniform circular motion onto one axis is sinusoidal.

Let

$$x(t) = R\cos(\omega t).$$

Then

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -R\omega\sin(\omega t),$$

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -R\omega^2\cos(\omega t).$$

Since $x(t) = R\cos(\omega t)$,

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\omega^2 x.$$

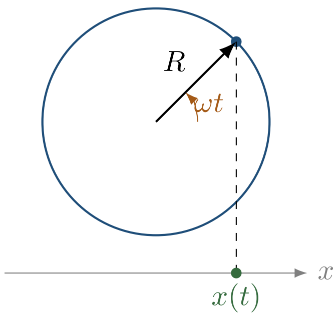
Therefore

$$\boxed{x'' + \omega^2 x = 0}.$$

This is the differential equation of simple harmonic motion.

26.1 Physical meaning

The one-dimensional projection of circular motion behaves like an oscillator. This is why sine and cosine appear in both circular motion and harmonic motion. The connection is not accidental; it comes from the geometry of rotation.



This section also shows a broader lesson: a single mathematical object may have several physical interpretations. The sine function can describe a coordinate of circular motion, a mass on a spring, a wave displacement, or an alternating voltage. The formula is not the physics; the interpretation makes it physics.

27 Angular momentum in circular motion

Angular momentum gives yet another language for circular motion. Linear momentum describes motion along a line. Angular momentum describes how much rotational motion a particle has about a chosen point. For a circular orbit, it takes a particularly simple form.

For a particle of mass m moving in a circle of radius r with speed v, the angular momentum magnitude about the centre is

$$L = mvr.$$

If $v = \omega r$, then

$$L = m\omega r^2.$$

For a point mass, the moment of inertia about the centre is

$$I = mr^2.$$

Thus

$$L = I\omega.$$

27.1 Why angular momentum matters

In central-force motion, the force points along the radius. The torque about the centre is

$$\tau = \mathbf{r} \times \mathbf{F}.$$

If $\mathbf{F}$ is parallel or antiparallel to $\mathbf{r}$, then

$$\tau = \mathbf{0}.$$

Therefore angular momentum is conserved.

This is deeper than the special case of circular motion. In planetary motion, conservation of angular momentum is connected with Kepler's second law. In the Bohr model, angular momentum quantization is the additional rule that selects allowed circular orbits.

28 Circular motion in the language of differential equations

We now return to the most compact mathematical description. The equation below says that acceleration is proportional to displacement and opposite in direction. This is the vector form of a restoring acceleration toward the centre. It is powerful because it describes not just one instant, but the entire structure of the motion.

Uniform circular motion can be written as a vector differential equation:

$$\mathbf{a} = -\omega^2\mathbf{r}.$$

Since acceleration is the second derivative of position,

$$\frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2} = -\omega^2\mathbf{r}.$$

This is

$$\boxed{\mathbf{r}'' + \omega^2\mathbf{r} = \mathbf{0}}.$$

In components:

$$x'' + \omega^2 x = 0,$$
$$y'' + \omega^2 y = 0.$$

A circular solution is

$$x(t) = R\cos(\omega t), \qquad y(t) = R\sin(\omega t).$$

28.1 But not every solution is a circle

The equations

$$x'' + \omega^2 x = 0,$$
$$y'' + \omega^2 y = 0$$

allow more general solutions:

$$x(t) = A\cos(\omega t) + B\sin(\omega t),$$
$$y(t) = C\cos(\omega t) + D\sin(\omega t).$$

Depending on constants, the trajectory can be an ellipse, a line segment, or a circle. Circular motion requires special initial conditions: equal amplitudes in perpendicular directions and a phase difference of $\pi/2$.

This is a useful warning. The same differential equation can describe different geometric motions depending on initial data.

Equation versus trajectory

A differential equation gives a family of possible motions. Initial conditions select one member of that family. Circular motion is not only about the equation; it is also about the initial conditions and constraints.

29 Dimensional checks

Before trusting a result, we should ask whether its units make sense. Dimensional analysis will not prove that a formula is correct, but it can quickly reveal many impossible results. It is also a way to read an equation physically: each side must describe the same kind of quantity.

Dimensional analysis is a simple way to detect many errors.

For centripetal acceleration:

$$a_c = \frac{v^2}{R}.$$

The dimension is

$$\frac{(\mathrm{m/s})^2}{\mathrm{m}} = \mathrm{m/s^2},$$

which is correct for acceleration.

For angular velocity:

$$\omega = \frac{2\pi}{T}.$$

Radians are dimensionless in SI, so the unit is

$$\mathrm{s^{-1}}.$$

For gravitational orbital speed:

$$v = \sqrt{\frac{GM}{r}}.$$

Since

$$[G] = \mathrm{m^3\,kg^{-1}\,s^{-2}},$$

we have

$$\left[\frac{GM}{r}\right] = \frac{\mathrm{m^3\,kg^{-1}\,s^{-2}\cdot kg}}{\mathrm{m}} = \mathrm{m^2/s^2}.$$

The square root gives m/s, as required.

30 A map of common student mistakes

The mistakes below are not random. Most of them come from treating equations as isolated formulas instead of as statements about a model. Reading the mistake carefully is therefore part of learning the method.

30.1 Mistake 1: treating centripetal force as an extra force

Incorrect:

$$F_g + F_c = ma.$$

Correct in a circular gravitational orbit:

$$F_g = m\frac{v^2}{r}.$$

The gravitational force is the centripetal force in the sense that it supplies the required inward net force.

30.2 Mistake 2: confusing speed with angular speed

Linear speed v and angular speed ω are related but not identical:

$$v = \omega R.$$

Two points on the same rotating disk have the same angular speed but different linear speeds if their radii are different.

30.3 Mistake 3: forgetting the direction of acceleration

For uniform circular motion, acceleration is not in the direction of motion. It is perpendicular to the velocity and points inward.

30.4 Mistake 4: using v^2/R with the wrong radius

The radius in

$$a_c = \frac{v^2}{R}$$

is the radius of the actual circular path of the body. In a banked curve, conical pendulum, or rotating platform, this may not be equal to the length of a string or the height of a cone. One must identify the geometry first.

30.5 Mistake 5: ignoring conditions of validity

The formula

$$r = \frac{mv}{|q|B}$$

for magnetic circular motion assumes that the velocity is perpendicular to a uniform magnetic field and that relativistic effects are negligible. If the velocity has a parallel component, the motion is helical. If the speed is close to the speed of light, the nonrelativistic formula must be modified.

31 One-page conceptual summary

Circular motion begins with geometry:

$$|\mathbf{r}| = R.$$

Differentiating gives the tangent condition:

$$\mathbf{r} \cdot \mathbf{v} = 0.$$

Uniform circular motion can be parametrised by

$$\mathbf{r}(t) = R\cos(\omega t)\mathbf{i} + R\sin(\omega t)\mathbf{j}.$$

Velocity is tangent:

$$\mathbf{v}(t) = -R\omega\sin(\omega t)\mathbf{i} + R\omega\cos(\omega t)\mathbf{j}.$$

Acceleration is inward:

$$\mathbf{a}(t) = -\omega^2\mathbf{r}(t).$$

Its magnitude is

$$a_c = \omega^2 R = \frac{v^2}{R}.$$

Newton's second law gives the central modelling condition:

$$\sum F_{\text{inward}} = m\frac{v^2}{R} = m\omega^2 R.$$

Every application asks: what real force supplies this inward resultant?

32 Practice problems with guided solutions

The solutions below are intentionally guided. The goal is not only to arrive at a number, but to show the sequence of decisions: identify the path, identify the inward direction, identify the real force, translate the condition, solve, and interpret.

These problems are not meant to be a formula drill. In each problem, first identify the physical condition and only then calculate.

Problem 1: rotating platform

A small object rests on a horizontal rotating platform at distance $R = 0.80\,\mathrm{m}$ from the centre. The coefficient of static friction is $\mu_s = 0.40$. What is the maximum angular speed before slipping?

Solution. The required inward force is static friction:

$$m\omega^2 R \leq \mu_s mg.$$

Cancel m:

$$\omega^2 \leq \frac{\mu_s g}{R}.$$

Thus

$$\omega_{\max} = \sqrt{\frac{0.40 \cdot 9.81}{0.80}} \approx 2.21\,\mathrm{s}^{-1}.$$

Problem 2: satellite speed

A satellite moves in a circular orbit of radius r around Earth. Express its speed in terms of G, Earth's mass M_E, and r.

Solution. Gravity supplies the inward force:

$$G\frac{M_E m}{r^2} = m\frac{v^2}{r}.$$

Cancel m and solve:

$$v = \sqrt{\frac{GM_E}{r}}.$$

Problem 3: magnetic radius

An electron enters a uniform magnetic field with speed v perpendicular to the field. What is the radius of its circular path?

Solution. The magnetic Lorentz force supplies the centripetal force:

$$evB = m_e\frac{v^2}{r}.$$

Cancel v:

$$r = \frac{m_e v}{eB}.$$

The direction of rotation depends on the electron's negative charge.

Problem 4: Bohr radius dependence

Using the Bohr conditions, show that the allowed radii satisfy $r_n \propto n^2$.

Solution. Use

$$\frac{ke^2}{r^2} = m_e \frac{v^2}{r}$$

and

$$m_e v r = n\hbar.$$

From the second equation,

$$v = \frac{n\hbar}{m_e r}.$$

Substitute into the first:

$$\frac{ke^2}{r^2} = \frac{n^2\hbar^2}{m_e r^3}.$$

Thus

$$r_n = \frac{n^2\hbar^2}{m_e k e^2}.$$

Therefore $r_n \propto n^2$.

33 Final synthesis: one structure, many worlds

The purpose of the document is not to make students memorize more formulas. The purpose is to make them see one recurring structure under many physical appearances. Once that structure is recognized, a problem about a car, a satellite, a charge, or a Bohr orbit stops being a new disconnected topic.

Circular motion is a meeting point of geometry, calculus, and dynamics.

Geometry gives the circle:

$$|\mathbf{r}| = R.$$

Calculus gives the acceleration:

$$\mathbf{a} = -\omega^2 \mathbf{r}.$$

Dynamics gives the force condition:

$$\sum F_{\text{inward}} = m\frac{v^2}{R}.$$

Different physical systems enter by changing the left-hand side:

$$T, \quad f_s, \quad G\frac{Mm}{r^2}, \quad k\frac{q_1 q_2}{r^2}, \quad |q|vB, \quad \ldots$$

The right-hand side is the kinematic requirement of circular motion. The left-hand side is the physical mechanism that makes such motion possible.

This is the main lesson:

A formula is not the beginning of understanding. It is the final compressed form of a modelling decision.

When solving circular-motion problems, do not begin by asking which formula to use. Begin by asking:

1. What is moving in a circle?
2. What is the radius of that circle?
3. Is the speed constant or changing?
4. What real forces act on the body?
5. Which component of the net force points toward the centre?
6. What relation follows from Newton's second law?
7. What does the result mean physically?

If these questions are clear, the formulas stop being magic. They become consequences.